HackInnovision INNOVISION 2026:

The hackathon begins at 12:00 AM (midnight) on November 14, 2026, and runs for 12 hours until 12:00 PM (noon).

1. THE THREE MISSION TRACKS

TRACK A:

PLANETARY RECONNAISSANCE & AUTONOMOUS NAVIGATION (ODYSSEY)

Theme: Space Technology, Computer Vision, and Uncertainty-Aware Exploration

The Scenario

An autonomous rover (MARS-09) is deployed in an unexplored, high-risk region of Mars. Operating under extreme communication delays and limited onboard compute, the rover cannot rely on mission control for micro-decisions. It must ingest raw planetary imagery, Digital Elevation Models (DEMs), slope measurements, and scientific objectives to map its environment and chart a safe, scientifically valuable path.

Key Technical Objectives

Planetary Perception & Terrain Intelligence: Process multi-spectral images and elevation point clouds to classify terrain, identify geological hazards, and construct 3D elevation maps.

Autonomous Path Planning & Cost Mapping: Implement robust navigation algorithms (e.g., A*, Dijkstra, RRT) to determine traversable corridors, compute movement costs, and avoid hazards.

Scientific Discovery & Uncertainty Reasoning: Rank candidate geological formations for exploration, identify potential resources (e.g., subsurface water ice), and quantify uncertainty in sensor readings and model predictions.

TRACK B:

AGENTIC WORKFLOW AUTOMATION & CONTEXT ENGINEERING (SECOND BRAIN)

Theme: Multi-Agent Orchestration, Tool Calling, and Reliable Execution

The Scenario

A major college event organizing team is managing complex workflows spread across email threads, messaging applications, calendars, and documents. When a key sponsor updates their financial commitment mid-stream, the change triggers cascading impacts across budgets, sponsorship trackers, promotional deliverables, and pending approvals.

Key Technical Objectives

- **Context Engineering & Knowledge Graphs:** Maintain persistent short- and long-term memory, extract entities (sponsors, deadlines, commitments), and resolve cross-application dependencies.
- **Specialized Agent Ecosystem:** Orchestrate collaborative agents (Communication Agent, Knowledge Agent, Planning Agent, Execution Agent, Verification Agent, and Coordinator Agent).
- **Reliable Execution & Adaptive Replanning:** Handle tool failures gracefully, implement idempotency and verification checks, and require human approval for high-consequence actions.

TRACK C:

CYBER INVESTIGATION & ATTACK GRAPH RECONSTRUCTION (CHIMERA)

Theme: Evidence-Driven Cyber Reasoning, Temporal Anomaly Detection & Explainable AI

The Scenario

Inside an enterprise network, malicious activity is unfolding silently. An employee account exhibits unusual login patterns, a service account gains elevated privileges, and an external connection opens in the noise. No single alarm triggers an alert; each event appears benign in isolation.

Key Technical Objectives

- **Attack Graph Reconstruction:** Model complex relationships between users, devices, services, and authentication events to map multi-stage lateral movement.
- **Temporal Reasoning & Anomaly Detection:** Correlate delayed, out-of-order, and fragmented logs to distinguish malicious signal from background noise.
- **Explainable AI & Root-Cause Analysis:** Trace conclusions directly back to underlying forensic evidence, identify initial access vectors, and estimate the potential blast radius.

2. 12-HOUR BUILD BOUNDARY & METHODOLOGY

Timeframe:	12-hour intensive build window.
Scope:	Teams are not expected to build production-grade enterprise platforms. Focus on a well-designed, end-to-end working prototype for one selected track (or an integrated cross-domain pipeline).
Tooling & Stack:	Use prepared datasets, simulated environments, public APIs, pretrained models, and modern agent frameworks. The emphasis is on architectural soundness, clean algorithmic reasoning, and a convincing end-to-end demonstration.

12-Hour Hackathon Timetable

TIME	ACTIVITY	DETAILS
12:00 AM – 2:30 AM	Hacking Session 1	Welcome address, rules, problem statements and judging criteria
2:30 AM – 04:00 AM	Mentor Round 1	Idea development, planning and initial implementation
04:00 AM – 04:30 AM	Refreshment Break	Tea, snacks and refreshments
04:30 AM – 07:30 AM	Hacking Session 2	Core development, feature implementation and integration
07:30 AM – 08:30 AM	Mentor Round 2	Technical guidance, progress review and feedback from mentors
08:30 AM – 09:30 AM	Final Hacking Sprint	Testing, debugging and project polishing
09:30 AM – 10:00 AM	Final Submission & Shortlisting	Teams submit their projects; jury selects the top 10 teams
10:00 AM – 12:00 PM	Top 10 Final Presentations	Each team gets 12 minutes: 8 minutes to present and 4 minutes for Q&A

3. EVALUATION CRITERIA

EVALUATION CRITERION	CORE FOCUS & WHAT JUDGES LOOK FOR
Technical Depth & Sophistication	Rigor of algorithms, model architecture, multi-agent orchestration, and system design.
Investigation & Reasoning Quality	Ability to connect fragmented clues into a coherent, defensible, and logical narrative.
Uncertainty & Robustness Handling	Resilience against noisy, missing, or conflicting data; clear communication of confidence limits.
Execution & Explainability	Traceability of conclusions/actions back to concrete evidence; safe, verified execution.
Innovation & Usability	Originality of approach, interface clarity, and effectiveness of the live demonstration.

4. RULES & GUIDELINES

- Teams must choose exactly one mission track to work on.
- All work and prototypes must be submitted strictly before the 12:00 PM (noon) deadline.
- All code and prototypes must be developed entirely during the 12-hour hackathon window (the use of open-source libraries, frameworks, and public APIs is permitted).
- Plagiarism or submitting pre-existing proprietary work will result in immediate disqualification.
- Participants must adhere to a professional code of conduct , harassment or disruptive behavior will not be tolerated.
- All decisions made by the judging panel are final and binding.

5. DELIVERABLES

Deliverables are intentionally kept flexible. Treat the items below as a guide rather than a checklist: submit whatever best shows your prototype, and choose the format, tooling and depth that suit your idea.

Core deliverables (all tracks)

- **Working prototype:** An end-to-end demo for your chosen track. A simulated or prepared dataset is fine.
- **Source code:** A repository link or zip, built during the 12-hour window.
- **Short write-up:** A README or one-pager covering your approach, architecture, assumptions and known limitations.
- **Final presentation:** If shortlisted, a live demo and walkthrough within the 8 + 4 minute slot.

Track-specific ideas (optional)

TRACK	WHAT YOU MIGHT SHOW
Track A: Odyssey	A terrain or cost map, a planned path with hazards avoided, and a ranked list of science targets with uncertainty shown.
Track B: Second Brain	A run-through of the sponsor change flowing across agents, including a recovered tool failure, a human approval step and an action log.
Track C: Chimera	A reconstructed attack graph with a timeline, each conclusion linked to its log evidence, plus the likely entry point and blast radius.

Whatever you submit, make it easy for judges to see the evaluation criteria at work: clear reasoning, honest handling of uncertainty, and traceable results.